1

1.1

We shift the coordinate system such that the string is tied at $x = 0, 2L$

$$k_n = \frac{\pi n}{2L}$$

$$u_x(x,t) = \frac{1}{\sqrt{L}} \sin\left(\frac{\pi n}{2L} x\right)$$

1.2

1.3

The wave is a superposition of the basis functions:

$$y(x,t) = \frac{1}{\sqrt{L}} \sum_{n=1}^{\infty} c_n \sin(\omega_n t) \sin(k_n x) \tag{1}$$

$$\omega_n = ck_n = c\frac{\pi n}{2L}$$

$$c = \sqrt{\frac{T}{\mu}}$$

Time derivative of :

$$\dot{y}(x,t) = \frac{1}{\sqrt{L}} \sum_{n}^{\infty} c_n \omega_n \cos(\omega_n t) \sin(k_n x)$$

$$\dot{y}(x,0) = \frac{1}{\sqrt{L}} \sum_{n}^{\infty} c_n \omega_n \sin(k_n x)$$

Inner product with basis function m:s

$$\int_0^{2L} \dot{y}(x,0)\, u_m(x)\, dx = \frac{1}{L} \sum_{n}^{\infty} c_n \omega_n \underbrace{\int_0^{2L} \sin(k_n x) \sin(k_m x)\, dx}_{L\delta_{n,m}}$$

$$\int_0^{2L} \dot{y}(x,0)\, u_n(x)\, dx = c_n \omega_n$$

$$\boxed{c_n = \frac{\int_0^{2L} \dot{y}\,(x,0) \sin\left(\frac{\pi n}{2L}x\right) dx}{\sqrt{L}\omega_n}} \tag{2}$$

Initial conditions:

$$\dot{y}\,(x,0) = \begin{cases} v_0 & x \in [L-a, L) \\ -v_0 & x \in [L, L+a] \\ 0 & otherwise \end{cases}$$

Into :

$$c_n = \frac{\int_0^{2L} \dot{y}\,(x,0) \sin\left(\frac{\pi n}{2L}x\right) dx}{\sqrt{L}\omega_n} = \frac{v_0}{\omega_n\sqrt{L}} \left\{ \int_{L-a}^{L} \sin\left(\frac{\pi n}{2L}x\right) dx - \int_{L}^{L+a} \sin\left(\frac{\pi n}{2L}x\right) dx \right\}$$

$$= \frac{v_0}{\omega_n\sqrt{L}} \frac{2L}{\pi n} \left\{ \left[-\cos\left(\frac{\pi n}{2}\right) + \cos\left(\frac{\pi n}{2L}(L-a)\right) \right] - \left[-\cos\left(\frac{\pi n}{2}(L+a)\right) + \cos\left(\frac{\pi n}{2}\right) \right] \right\}$$

$$= \frac{v_0}{\omega_n\sqrt{L}} \frac{2L}{\pi n} \left\{ -2\cos\left(\frac{\pi n}{2}\right) + \underbrace{\cos\left(\frac{\pi n}{2L}(L-a)\right)}_{\cos\left(\frac{\pi n}{2}\right)\cos\left(\frac{\pi n}{2}\frac{a}{L}\right)+\sin\left(\frac{\pi n}{2}\right)\sin\left(\frac{\pi n}{2}\frac{a}{L}\right)} + \underbrace{\cos\left(\frac{\pi n}{2L}(L+a)\right)}_{\cos\left(\frac{\pi n}{2}\right)\cos\left(\frac{\pi n}{2}\frac{a}{L}\right)-\sin\left(\frac{\pi n}{2}\right)\sin\left(\frac{\pi n}{2}\frac{a}{L}\right)} \right\}$$

$$= \frac{v_0}{\omega_n\sqrt{L}} \frac{2L}{\pi n} \left\{ -2\cos\left(\frac{\pi n}{2}\right) + \cos\left(\frac{\pi n}{2}\right)\cos\left(\frac{\pi n}{2}\frac{a}{L}\right) + \sin\left(\frac{\pi n}{2}\right)\cancel{\sin\left(\frac{\pi n}{2}\frac{a}{L}\right)} + \cos\left(\frac{\pi n}{2}\right)\cos\left(\frac{\pi n}{2}\frac{a}{L}\right) - \cancel{\sin\left(\frac{\pi n}{2}\right.} \right.$$

$$= \frac{2v_0}{\omega_n\sqrt{L}} \frac{2L}{\pi n} \cos\left(\frac{\pi n}{2}\right) \left\{ \cos\left(\frac{\pi n}{2}\frac{a}{L}\right) - 1 \right\}$$

$$\boxed{\boxed{c_n = \frac{2}{\sqrt{L}} \frac{v_0}{c} \left(\frac{2L}{\pi n}\right)^2 \cos\left(\frac{\pi n}{2}\right) \left\{ \cos\left(\frac{\pi n}{2}\frac{a}{L}\right) - 1 \right\}}}$$

$$c_1 = \frac{2}{\sqrt{L}} \frac{v_0}{c} \left(\frac{2L}{\pi}\right)^2 \cancel{\cos\left(\frac{\pi}{2}\right)} \left\{ \cos\left(\frac{\pi}{2}\frac{a}{L}\right) - 1 \right\} = 0$$

$$c_2 = \frac{2}{\sqrt{L}} \frac{v_0}{c} \left(\frac{L}{\pi}\right)^2 \left\{ 1 - \cos\left(\pi\frac{a}{L}\right) \right\}$$

$$c_3 = \frac{2}{\sqrt{L}} \frac{v_0}{c} \left(\frac{2L}{3\pi}\right)^2 \cancel{\cos\left(\frac{3\pi}{2}\right)} \left\{ \cos\left(\frac{3\pi}{2}\frac{a}{L}\right) - 1 \right\} = 0$$

Since the function is antisymmetric around $x = L$ all the symmetric modes (around $x = L$ must vanish).

1.4

$$t = L\sqrt{\frac{\mu}{T}} = \frac{L}{c}$$

$$y(x,t) = \frac{1}{\sqrt{L}} \sum_{n=1}^{\infty} c_n \sin\left(ck_n \frac{L}{c}\right) \sin(k_n x)$$

Since we want to keep only the odd functions (i.e. even values of n):

$$n = 2m$$

$$y\left(x, L\sqrt{\frac{\mu}{T}}\right) = \frac{1}{\sqrt{L}} \sum_{m=0}^{\infty} c_m \sin(k_m L) \sin(k_m x)$$

$$= \frac{1}{\sqrt{L}} \sum_{m=0}^{\infty} c_m \underbrace{\sin(\pi n)}_{0} \sin(k_m x) = 0$$

2

2.1

Initial field:

$$y = A \sin(kx - \omega t)$$

Intensity:

$$I_0 = \frac{c\epsilon_0}{2} |A|^2$$

After BS1

Transmitted:

$$y\left(0^+, t\right) = \frac{A}{\sqrt{2}} \sin(-\omega t)$$

Reflected:

$$y\left(0^{+},t\right)=\frac{A}{\sqrt{2}}\sin\left(\omega t+\frac{\pi}{2}\right)$$

After M1

$$y\left(L^{+},t\right)=\frac{A}{\sqrt{2}}\sin\left(kL-\omega t+\pi\right)$$

After M2

$$y\left(L^{+},t\right)=\frac{A}{\sqrt{2}}\sin\left(kL-\omega t+\frac{3\pi}{2}\right)$$

2.2

Field at A

$$y=\frac{1}{2}\sin\left(2kL-\omega t+\pi\right)+\frac{1}{2}\sin\left(2kL-\omega t+2\pi\right)=0$$

Intensity at A

$$I=0$$

Field at B

$$y=A\sin\left(2kL-\omega t+\frac{3\pi}{2}\right)$$

Intensity at B

$$I=I_0$$

2.3

A

$$y=\frac{A}{2}\sin\left(2k_0L-\omega t+\pi\right)+\frac{A}{2}\sin\left(k_0\left(2L-l\right)+k_0n_1l-\omega t\right)$$

$$=\frac{A}{2}\sin\left(\underbrace{2k_0L-\omega t}_{\equiv\phi_0}\right)+\frac{A}{2}\sin\left(2k_0L-\omega t+\underbrace{k_0\left(n_1-1\right)l}_{\equiv\Delta\phi}+\pi\right)$$

$$=\frac{A}{2}\left[\sin\left(\phi_0\right)+\sin\left(\phi_0+\Delta\phi+\pi\right)\right]=-A\cos\left(\frac{\Delta\phi+\pi}{2}\right)\sin\left(\frac{2\phi_0+\Delta\phi}{2}\right)$$

$$I=I_0\cos^2\left(\frac{\Delta\phi+\pi}{2}\right)=I_0\sin^2\left(\frac{\Delta\phi}{2}\right)=I_0\sin^2\left(\frac{k_0\left(n_1-1\right)l}{2}\right)$$

$$\boxed{I_A = I_0 \sin^2\left(\frac{k_0\left(n_1-1\right)l}{2}\right)}$$

B

$$\boxed{I_B = I_0 \cos^2\left(\frac{k_0\left(n_1-1\right)l}{2}\right)}$$

$$\boxed{I_A + I_B = I_0\left(\sin^2\left(\frac{k_0\left(n_1-1\right)l}{2}\right) + \cos^2\left(\frac{k_0\left(n_1-1\right)l}{2}\right)\right) = I_0}$$

2.4

A

$$y = \frac{A}{2}\sin\left(\underbrace{2k_0L-\omega t}_{\equiv\phi_0}\right) + \frac{A}{2}\sin\left(2k_0L-\omega t+\underbrace{k_0\left(n_1-1\right)l}_{\equiv\Delta\phi}+\pi\right)e^{-k_0\kappa l}$$

$$= \frac{A}{2}\left[\sin\left(\phi_0\right)+\sin\left(\phi_0+\Delta\phi+\pi\right)e^{-k_0\kappa l}\right]$$

$$= \frac{A}{2}\left[\sin\left(\phi_0\right)-\sin\left(\phi_0+\Delta\phi\right)e^{-k_0\kappa l}\right]$$

The lowest value that can be measured is at $\Delta\phi = 0$, but instead of measuring 0:

$$y = \frac{A}{2}\left[1-e^{-k_0\kappa l}\right]\sin\left(\phi_0\right)$$

$$I = I_0\frac{\left(1-e^{-k_0\kappa l}\right)^2}{4}$$

The highest value, at $\Delta\phi = \pi$:

$$I = I_0\frac{\left(1+e^{-k_0\kappa l}\right)^2}{4}$$